

The Price of De-correlation: Quantifying the Agreement–Cost Trade-off in Heterogeneous Large Language Model Ensembles

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Abstract

Ensembles of large language model (LLM) “agents” are increasingly deployed as committees that vote on decisions, on the premise that aggregating multiple opinions reduces error. This premise implicitly assumes the agents are *independent*; yet a homogeneous ensemble—several instances of one model—may behave as a near-echo-chamber whose members merely restate a shared prior. We ask, and quantify, a concrete question: how much does introducing *cross-provider model heterogeneity* reduce within-ensemble agreement, and at what inference-cost premium? Using a pre-registered design, we run 54,000 independent LLM calls across three five-agent ensemble configurations spanning heterogeneity levels $\{0, 1, 3\}$ (measured by the number of non-primary providers), evaluated on a directional financial-analysis task over 100 equities $\times$ 12 dates. Agreement is measured by Fleiss’ κ on per-cell directional calls, with cluster-bootstrap confidence intervals resampling equities. We report three findings. *First*, heterogeneity de-correlates ensembles substantially and monotonically: within-ensemble agreement falls from $\kappa = 0.552$ (homogeneous) to 0.302 (one heterogeneous agent) to 0.215 (three), a primary effect of $\Delta\kappa = 0.336$ (95% CI $[0.304, 0.369]$). *Second*, this de-correlation carries a $\sim 4.4\times$ inference-cost premium per ensemble decision, yielding an explicit agreement–cost frontier; a capability-matched control (efficient-tier models across providers) reproduces the effect ($\Delta\kappa = 0.315$, 95% CI $[0.288, 0.341]$, $\approx 94\%$ of the full effect), indicating it is driven by provider identity, not capability tier. *Third*, and as a methodological contribution, we show that a common measurement artifact—sampling non-determinism under a shared sampling context—inflates the apparent effect by roughly $4\times$ ($\Delta\kappa$ drops from 0.485 to 0.113 in a controlled pilot once each agent is given an independent draw); we provide a protocol that removes it. We find *no* reliable improvement in directional accuracy from heterogeneity at this scale, consistent with recent “algorithmic monoculture” cautions: diversity reduces surface agreement without, in our setting, purchasing average correctness. An exploratory analysis nonetheless finds that homogeneous ensembles produce confident, unanimous, and *wrong* directional

calls at roughly $1.8\times$ the heterogeneous rate, suggesting the de-correlation buys tail-risk reduction even where it buys no gain in mean accuracy. All artifacts (frozen dataset, seeded deterministic analysis) are released for reproduction.

Index Terms— *Large language models, multi-agent systems, ensemble methods, inter-rater agreement, Fleiss’ kappa, model diversity, algorithmic monoculture, reproducibility, computational finance.*

1 Introduction

Multi-agent LLM systems increasingly aggregate several model “agents” into a committee that debates or votes, on the wisdom-of-crowds premise that combining opinions cancels individual error [3, 2]. That premise is load-bearing on *independence*: crowd aggregation only helps to the extent that members err differently. A homogeneous ensemble— k instances of a single model queried independently—risks being a costly echo chamber, its members restating one shared prior. The natural remedy is *heterogeneity*: mixing models from different providers. But heterogeneity is not free, and its benefit is not guaranteed: recent work shows LLM errors are often strongly correlated even across providers and architectures, an “algorithmic monoculture” that limits the value of naive diversity [1].

The magnitude of that de-correlation is not a foregone conclusion; it is the subject of two *competing predictions* in the current literature. The naive-diversity view—implicit in the popularity of “mixture-of-agents” designs—holds that mixing models from different providers, trained on different data with different architectures, should buy substantial independence. The opposing view, argued directly by Kim *et al.* [1], is that LLM errors are strongly correlated *even across providers and architectures*—an “algorithmic monoculture”—so cross-provider mixing may de-correlate outputs only marginally. These predictions are quantitatively far apart: ex ante, the primary effect could plausibly have ranged from $\Delta\kappa \approx 0$ (monoculture dominates; the frontier is flat and heterogeneity is a waste of money) to a large positive value (diversity is

real and priceable). **Had provider identity not mattered, the agreement–cost frontier would have been flat**—so measuring its slope is a genuine test, not a restatement of intuition. We resolve it empirically: $\Delta\kappa = 0.336$, a large and reliably nonzero effect, but one that must be *paid for*.

Despite the popularity of heterogeneous designs, the quantity that governs whether they can help—how much heterogeneity actually *de-correlates* agent outputs—is rarely measured with statistical rigor, and its *cost* is rarely priced. Two obstacles recur. (i) *Estimand ambiguity*: agreement is often computed over agents that share a sampling context, so measured “agreement” partly reflects a measurement artifact rather than genuine model similarity. (ii) *No cost accounting*: papers report that diversity “helps” without quantifying the compute premium it demands. Our contribution is thus the *magnitude*, the *price*, and the *protocol*—none of which is derivable from the intuition that “different models disagree.”

Why quantifying this matters in practice: LLM committees are increasingly placed in high-stakes loops (financial, clinical, operational) precisely to gain the robustness of a “second opinion.” If a homogeneous committee is in fact an echo chamber, that robustness is illusory—the redundant calls waste compute and, worse, manufacture false confidence by turning one model’s opinion into an apparent consensus. A practitioner therefore needs to know both *how much* genuine independence heterogeneity buys and *what it costs*, so that the premium is paid only when the added independence is real. This paper supplies both numbers.

Contributions This paper makes the following contributions.

1. A **pre-registered, large-scale (54,000-call) measurement** of how cross-provider heterogeneity affects within-ensemble agreement in LLM committees, with cluster-bootstrap confidence intervals; we find a substantial, monotone de-correlation ($\Delta\kappa = 0.336$, 95% CI [0.304, 0.369]).
2. An explicit **agreement–cost frontier**: the $\sim 4.4\times$ inference-cost premium that buys this de-correlation, giving practitioners a concrete trade-off curve rather than a qualitative claim.
3. A **measurement protocol** for ensemble agreement that isolates independent-draw agreement from sampling non-determinism, and a demonstration that ignoring it inflates the effect $\sim 4\times$.
4. An **exploratory downstream analysis** showing that, although average directional accuracy does not improve, heterogeneity reshapes the *structure* of consensus: homogeneous committees commit confident, unanimous, directional errors at $1.84\times$ the

fully heterogeneous rate, suggesting the priced independence buys tail-risk reduction in this setting.

5. A **fully reproducible artifact**: a frozen, hash-verified dataset and a deterministic, seeded analysis pipeline that reproduces every figure and table.

We deliberately separate *agreement* from *accuracy* and report an honest null on the latter for this directional-finance task, positioning the work within the monoculture literature.

2 Related Work

2.1 LLM ensembles and mixture-of-agents

Aggregating multiple LLM outputs—by voting, debate, or layered mixture-of-agents—has become a standard technique for boosting reliability [3, 2]. A recurring empirical caveat is that naive aggregation yields marginal gains over the strongest single model and can amplify systematic error when a strong model is outvoted by correlated weaker ones [1]. Scaling studies further report diminishing or negative returns from adding *homogeneous* agents, while injecting *diversity* yields more sustained gains. This has motivated diversity-*optimized* ensembles that explicitly select or weight members to maximize a diversity criterion—e.g. the focal-diversity pruning of LLM-TOPLA [4] and the fingerprint-diversity ensemble DFPE [5]—and heterogeneous consensus frameworks such as Council Mode [7] that aggregate cross-family agents to curb hallucination. These methods *presuppose* that heterogeneity supplies independence and evaluate it through downstream accuracy; by contrast, we directly *measure* how much independence heterogeneity actually supplies, quantify its statistical significance, and price it.

2.2 Correlated errors and algorithmic monoculture

Kim *et al.* [1] document substantial error correlation across hundreds of LLMs, including across providers, and connect it to systemic risks of “algorithmic monoculture.” Their focus is error correlation (agreement conditional on being wrong); ours is output agreement on a decision task. The two are complementary: we show heterogeneity reduces surface agreement, yet—consistent with their caution—without a measurable accuracy benefit in our setting. Closely related and concurrent, Bugaud [6] shows in the vision-language modality that models of the same *architectural family* share correlated errors that shrink the effective number of independent voters, and that family-aware aggregation recovers accuracy. Our Fig. 4 is the text-LLM, provider-level analogue—same-source agents agree far more than cross-source ones—obtained here through a pre-registered,

statistically powered measurement with an attached cost frontier.

2.3 Inter-rater agreement and diversity metrics

Fleiss’ κ [8] is the standard chance-corrected agreement statistic for multiple raters on categorical labels; we adopt it, treating agents as interchangeable raters and (ticker, date) cells as subjects. Prior ensemble-selection work uses focal-diversity [4] and information-theoretic criteria to prune correlated members; we instead treat agreement as the object of measurement and quantify its dependence on provider heterogeneity and cost, with cluster-bootstrap confidence intervals.

2.4 Reproducibility and pre-registration

Pre-registration and frozen, hash-verified datasets remain uncommon in empirical LLM studies. We adopt a confirmatory design—freeze the plan, collect once, re-analyze deterministically—to reduce researcher degrees of freedom, and we release the artifacts.

3 Problem Formulation and Metrics

3.1 Ensemble configurations

Each ensemble comprises five agents that independently answer the same prompt. We define three configurations by *heterogeneity level* h , the count of non-primary-provider agents:

- **HOM** ($h=0$): $5 \times$ primary model (OpenAI efficient tier);
- **HET-LITE** ($h=1$): $4 \times$ primary + $1 \times$ Google flash;
- **HET** ($h=3$): $2 \times$ primary + $2 \times$ Google flash + $1 \times$ xAI flagship.

Exact model strings and pricing appear in Table 1.

3.2 Agreement statistic

For a configuration c and cell $s = (\text{ticker}, \text{date})$, the five agents emit directional labels in $\{\text{BUY}, \text{HOLD}, \text{SELL}\}$. We compute Fleiss’ κ over cells within each run and average across runs, giving κ_c . The pre-registered *primary endpoint* is

$$\Delta\kappa = \kappa_{\text{HOM}} - \kappa_{\text{HET}}, \quad (1)$$

with pre-registered secondary contrasts $\kappa_{\text{HOM}} - \kappa_{\text{HET-LITE}}$ and $\kappa_{\text{HET-LITE}} - \kappa_{\text{HET}}$.

3.3 Independent-draw estimand

The quantity of interest is agreement among *independent* agent draws. We therefore issue one fresh application programming interface (API) call per agent (no shared context, no cached completions) and assign each agent a unique, reproducible seed $\text{seed} = \text{SHA-256}(\text{master}, \text{run}, c, \text{ticker}, \text{date}, \text{agent}) \bmod 2^{31}$. As a diagnostic we compare *within-run* κ (five agents under one sampling context) against *cross-run* κ (pooled draws across runs); a large positive gap signals that shared-context sampling non-determinism is inflating measured agreement (Section 5.4).

3.4 Uncertainty quantification

Confidence intervals (CIs) use a *cluster bootstrap* that re-samples equities (the clustering unit), 2,000 draws, fixed random-number-generator (RNG) seed 42, percentile method. A verdict of CONFIRMED requires $\Delta\kappa > 0$ with a 95% CI excluding zero.

4 Experimental Design

4.1 Task and grid

Each agent receives a dated, no-lookahead context snippet for one equity and returns strict JavaScript Object Notation (JSON): `{"direction", "conviction", "rationale"}`. The grid is 100 large-cap equities stratified across all 11 Global Industry Classification Standard (GICS) sectors $\times$ 12 analysis dates in 2026×3 configurations $\times$ 3 runs $\times$ 5 agents = 54,000 calls. Dates are chosen strictly after each model’s knowledge cutoff and early enough that a 20 trading-day forward-return window has closed (for the accuracy proxy), and are spread across distinct 2026 market regimes.

4.2 No-lookahead context

Context snippets are *price-derived*: for each cell we compute, from the equity’s own price history up to the trading day *before* the analysis date, its trailing 1-week/1-month/3-month returns, 21-day realized volatility, and position within its 63-day range. This uses only public prices, contains no post-date information, and is fully reproducible; provenance is logged per call.

4.3 Models

All agents run at temperature 0.7. Exact model strings are pinned; because providers update models, the frozen dataset—not the live endpoints—is the object of record.

Table 1: Model registry (verified via a live probe). Prices are USD per 10^6 tokens at time of experiment.

Role	Model string	In	Out
Primary / efficient (OpenAI)	<code>gpt-5.4-nano</code>	0.20	1.25
Heterogeneous (Google)	<code>gemini-3.5-flash</code>	1.50	9.00
Heterogeneous (xAI)	<code>grok-4.3</code>	1.25	9.00

Table 2: Primary and secondary $\Delta\kappa$ (cluster bootstrap over equities, 2,000 draws, seed 42).

Contrast	$\Delta\kappa$	95% CI
HOM – HET (primary)	0.336	[0.304, 0.369]
HOM – HET-LITE	0.249	[0.218, 0.280]
HET-LITE – HET	0.087	[0.067, 0.105]

4.4 Pre-registration, freezing, and reproducibility

The design (endpoints, grid, seeds, prompt, decision rule) was frozen and SHA-256-hashed before confirmatory data collection. Raw data were collected once into an immutable, read-only, hash-stamped `runs.csv`; the analysis reads only that file plus the frozen configuration, with all randomness seeded, and reproduces every figure and table byte-for-byte. A power analysis on an independent pilot sized the grid.

5 Results

Analysis is over the complete grid (54,000/54,000 cells).

5.1 Heterogeneity de-correlates ensembles (primary)

Within-ensemble agreement decreases sharply and *monotonically* with heterogeneity (Table 4, Fig. 1): $\kappa_{\text{HOM}} = 0.552$, $\kappa_{\text{HET-LITE}} = 0.302$, $\kappa_{\text{HET}} = 0.215$. The pre-registered primary effect is

$$\Delta\kappa(\text{HOM-HET}) = 0.336, \quad 95\% \text{ CI } [0.304, 0.369],$$

whose interval excludes zero: verdict **CONFIRMED**. In Landis-Koch terms, homogeneous ensembles reach “moderate-substantial” internal agreement while fully heterogeneous ones fall to “fair,” i.e. markedly more independent.

Both secondary contrasts are also positive with CIs excluding zero (Table 2), confirming that *each* increment of heterogeneity contributes de-correlation.

5.2 Robustness of the agreement statistic

Fleiss’ κ is prevalence-sensitive (the well-known “ κ paradox”), and our BUY/HOLD/SELL marginals do shift

Table 3: Agreement under three statistics (per-run averaged), with the per-config directional marginals. All three statistics agree on the ordering despite the prevalence shift in HOLD.

Config	agreement			marginals		
	κ	α	AC1	BUY	HOLD	SELL
HOM	0.552	0.552	0.743	0.212	0.709	0.079
HET-LITE	0.302	0.303	0.548	0.282	0.653	0.064
HET	0.215	0.216	0.398	0.358	0.556	0.085

Table 4: Frontier: within-ensemble agreement and cost by heterogeneity level.

Config	h	κ	\$/ensemble-decision
HOM	0	0.552	6.48×10^{-4}
HET-LITE	1	0.302	1.52×10^{-3}
HET	3	0.215	2.84×10^{-3}

with heterogeneity—the HOLD share falls from 0.71 (HOM) to 0.56 (HET) as agents grow more willing to take directional stances (Table 3). To confirm the ordering is not an artifact of this prevalence shift, we recompute agreement with two statistics that treat chance agreement differently: Krippendorff’s α (nominal) and Gwet’s AC1, which is specifically designed to be robust to the paradox. All three rank the configurations identically and monotonically, $\text{HOM} > \text{HET-LITE} > \text{HET}$ (Table 3); α tracks κ to three decimals, and while AC1’s absolute values are higher (as expected), it preserves the ordering and the large HOM→HET gap. The de-correlation result is therefore not an artifact of the chosen agreement statistic.

5.3 The agreement–cost frontier

De-correlation is not free. Cost per ensemble decision rises from $\$6.5 \times 10^{-4}$ (HOM) to $\$1.5 \times 10^{-3}$ (HET-LITE) to $\$2.8 \times 10^{-3}$ (HET)—a $\sim 4.4\times$ premium from monoculture to full heterogeneity (Table 4). Fig. 1 plots the resulting frontier: agreement (lower is more diverse) against cost, a curve practitioners can use to select an operating point.

5.4 Measurement protocol: sampling non-determinism inflates κ

Exactly what was shared In our first (pilot) implementation the random seed was derived *per ensemble run* rather than *per agent*: every one of the five same-model agents in a cell received the *identical* seed, and therefore the identical sampling trajectory, from the provider’s decoder. With temperature sampling made deterministic by a shared seed, five “independent” `gpt-5.4-nano` calls on the same prompt returned near-

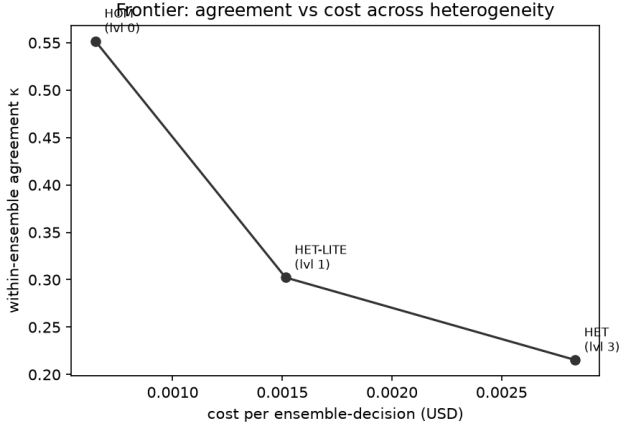


Figure 1: Agreement-cost frontier. Within-ensemble Fleiss’ κ (y) versus cost per ensemble decision (x) at heterogeneity levels $\{0, 1, 3\}$. Lower κ = more independent agents; the horizontal spread is the cost premium.

Before (artifact): one seed shared by all agents in a run
`seed = f(master, run)`
for agent in 1..5: `call(model, prompt, seed)` #
identical draw
After (fix): unique per-agent seed, independent draws
`seed = SHA256(master, run, config,`
`ticker, date, agent) & 0x7FFFFFFF`
`call(model, prompt, seed)` # independent draw per agent

Figure 2: The one-line fix. The pilot derived the seed per run, so all five same-model agents shared a sampling path; the study derives a distinct integer-safe seed per agent, restoring independent draws from the same model.

verbatim outputs, driving κ_{HOM} artificially toward 1. The fix is one line of seed derivation: give each agent a unique, reproducible seed keyed by its position (Fig. 2), so the five agents draw independent samples from the *same* model. We stress that this is the *only* change; the prompt, model, temperature, and grid are identical before and after.

Ruling out adjacent explanations A reader may wonder whether the near-duplicate pilot outputs came instead from response caching or request batching. Neither applies: provider- and client-side prompt caching was disabled (no cache headers; each call is a fresh completion), and every agent call is an independent HTTP request—agents are never batched into a single multi-sample call. The duplication was attributable solely to the shared seed, which the fix removes.

Our within-run vs. cross-run diagnostic (Table 5) *verifies* the fix rather than merely asserting it: after adopting per-agent independent draws, the HOM within-cross gap is $0.0006 \approx 0$, confirming the homogeneous agents are genuinely independent rather than

Table 5: Non-determinism check: within-run vs. cross-run κ (must be ≈ 0 for the primary/homogeneous configuration).

Config	within	cross	within-cross
HOM	0.552	0.551	+0.001
HET-LITE	0.302	0.350	−0.048
HET	0.215	0.289	−0.074

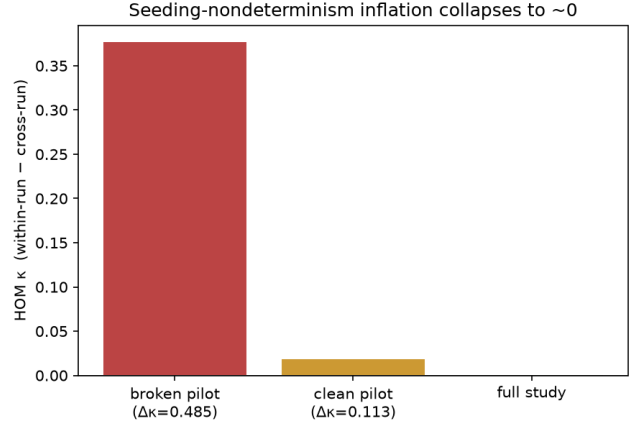


Figure 3: The measurement trap. Homogeneous within-cross κ gap collapses from a seed-shared pilot ($\Delta\kappa = 0.485$) to the independent-draw pilot ($\Delta\kappa = 0.113$) to ≈ 0 in the full study, showing that shared-context sampling non-determinism, not model similarity, produced the inflation.

copies (a residual gap near 1 would have signalled surviving duplication). In a controlled pilot, the artifact was severe: the same primary effect measured $\Delta\kappa = 0.485$ under a shared sampling context but 0.113 under independent draws—a $\sim 4\times$ over-statement produced entirely by the measurement apparatus (Fig. 3). We recommend the independent-draw protocol and the within/cross diagnostic as a standard check for any LLM-ensemble agreement study.

The negative within-cross gaps for the *heterogeneous* configurations (−0.048 and −0.074; Table 5) are expected and benign, and act as a further consistency check. Pooling the three runs’ draws into a single cross-run κ re-introduces *same-model replicate* agreement—a model’s repeated draws for a cell tend to be mutually similar—which raises the pooled κ above the single-draw within-run value; the more diverse the ensemble, the larger this pooling effect, hence the monotonic ordering of the gaps (0 for HOM, −0.048 for HET-LITE, −0.074 for HET). Crucially, our primary endpoint uses the *conservative* within-run (single-draw) estimate, so this pooling cannot inflate $\Delta\kappa$; the direction of the gaps merely reconfirms that additional same-model draws add correlation.

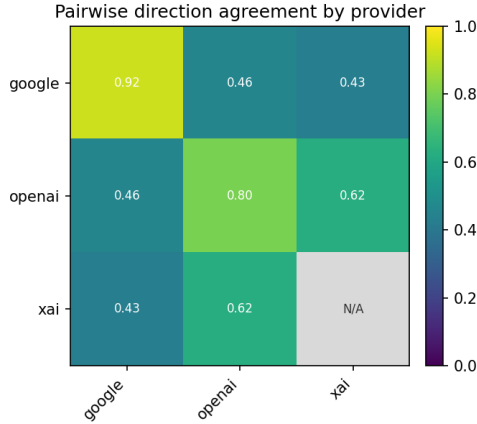


Figure 4: Pairwise directional-agreement matrix by provider. Diagonal (within-provider) agreement greatly exceeds off-diagonal (cross-provider), the source of ensemble de-correlation. The xAI-xAI diagonal is *undefined* (marked N/A, not 0): each HET cell contains a single xAI agent, so no within-xAI agent pair exists.

Table 6: Accuracy proxy (SECONDARY; underpowered by design). Hit-rate of majority-vote direction vs. 20-day forward-return sign; HOLD abstains.

Config	hit-rate	n	95% CI (Wilson)
HOM	0.438	315	[0.384, 0.493]
HET-LITE	0.463	339	[0.411, 0.516]
HET	0.488	525	[0.445, 0.530]

5.5 Provider-pair agreement

The pairwise direction-agreement matrix (Fig. 4) shows agents from the *same* provider agree far more than cross-provider pairs: within-provider agreement is 0.92 (Google) and 0.80 (OpenAI), whereas cross-provider pairs range 0.43–0.63. This is the mechanism behind the frontier: provider identity, not merely stochasticity, is the dominant axis of agent (dis)agreement.

5.6 Accuracy proxy (secondary; honest null)

As a pre-declared *secondary*, descriptive axis, we score each ensemble’s majority-vote direction against the sign of the realized 20 trading-day forward return (HOLD abstains). Directional hit-rates are 0.438 (HOM), 0.463 (HET-LITE), and 0.488 (HET), with overlapping 95% Wilson intervals (Table 6). The study is powered for κ , *not* for accuracy; we therefore draw *no* conclusion that heterogeneity improves accuracy. The slight upward trend is reported for completeness only.

5.7 Beyond average accuracy: manufactured consensus and tail risk (exploratory)

Average directional accuracy is flat (Sec. 5.6), but average accuracy is not the only quantity a practitioner cares about: in a decision setting, a committee that is *confidently and unanimously wrong* is more dangerous than one whose members hedge. We therefore examine, *post hoc and exploratorily* (no pre-registered hypothesis; no confidence intervals promoted to our headline claims), how heterogeneity reshapes the *structure* of consensus rather than its mean correctness. All quantities below are computed on the same frozen grid, scoring each five-agent ensemble decision against the realized forward-return sign (Table 7).

Three patterns emerge, all monotone in heterogeneity. (i) *Unanimity*. The rate at which all five agents return one identical direction falls from 0.591 (HOM) to 0.315 (HET-LITE) to 0.267 (HET): homogeneous committees manufacture consensus more than twice as often. (ii) *Confident collective error*. The rate of decisions that are unanimous, directional, *and wrong*—the exact “manufactured-consensus” failure that monoculture warnings describe [1]—is 0.058 for HOM versus 0.032 for HET, so the homogeneous ensemble commits confident collective blunders at $1.84\times$ the fully heterogeneous rate. (iii) *Error pile-on*. Conditional on the majority vote being *wrong*, the fraction of agents that nonetheless endorse the wrong call is 0.824 (HOM), 0.761 (HET-LITE), 0.636 (HET): when a heterogeneous ensemble errs, its members are far more likely to have dissented, leaving a usable disagreement signal that a homogeneous ensemble erases. Read together, these indicate that the priced de-correlation buys a measurable reduction in *tail risk*—fewer confident collective errors—even where it buys no gain in mean accuracy, which for a finance application is arguably the more relevant benefit.

We caution against over-reading conviction as calibrated confidence: per-agent directional hit-rate is essentially flat across self-reported conviction buckets and never exceeds chance by a reliable margin in any configuration (Fig. 5), consistent with the accuracy null. Heterogeneity’s benefit here is in the *structure* of collective agreement, not in per-call calibration. These exploratory results motivate, but do not establish, a downstream advantage; we frame them as hypotheses for a confirmatory, appropriately powered follow-up.

5.8 Robustness: capability-matched control

To separate provider diversity from capability tier—the study’s principal confound—we ran a pre-registered control that holds capability approximately constant while varying provider: HET-SAMETIER, comprising three gpt-5.4-nano and two gemini-3.1-flash-lite

Table 7: Consensus structure by heterogeneity (exploratory; 3,600 scoreable five-agent decisions per config). “Unan. wrong” = all five agents agree on one directional call that is wrong; “pile-on” = $P(\text{agent endorses majority} \mid \text{majority wrong})$.

Config	unanimity	unan. wrong	pile-on
HOM	0.591	0.058	0.824
HET-LITE	0.315	0.030	0.761
HET	0.267	0.032	0.636

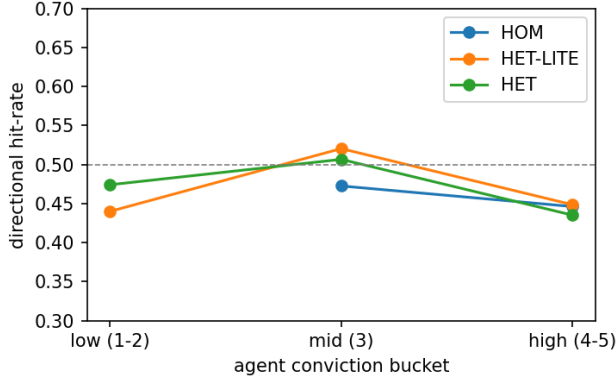


Figure 5: Conviction is not calibrated. Per-agent directional hit-rate by self-reported conviction bucket; no configuration rises reliably above chance (0.5, dashed) and higher conviction does not imply higher accuracy, consistent with the accuracy null (Sec. 5.6).

agents, i.e. the *efficient* tier of each provider (no efficient-tier xAI model is exposed, so the control is OpenAI vs. Google). Compared against the identical HOM baseline over the same 100×12 grid (36,000 calls), the control yields $\kappa_{\text{HOM}} = 0.552$ and $\kappa_{\text{HET-SameTier}} = 0.236$, a contrast of

$$\Delta\kappa(\text{HOM-HET-SameTier}) = 0.315, \quad 95\% \text{ CI } [0.288, 0.341]$$

which excludes zero. This capability-matched effect recovers $\approx 94\%$ of the full flagship-mixture effect (0.336), demonstrating that the de-correlation is driven overwhelmingly by *provider identity* rather than capability tier. The capability confound is therefore not a plausible explanation for the primary result. (The reproduction of $\kappa_{\text{HOM}} = 0.552$ to four decimals across two independently collected datasets also re-confirms determinism of the homogeneous baseline.)

5.9 Robustness: temperature sensitivity

Agreement is mechanically a function of sampling temperature, so a natural concern is that our primary effect is an artifact of the single operating point $T = 0.7$. We therefore re-ran a small stratified subgrid (8 equities $\times$ 2 dates $\times$ 3 configs $\times$ 3 runs, written to separate files that

Table 8: Temperature-sensitivity robustness (8×2 subgrid, 3 runs per cell). The ordering ($\kappa_{\text{HOM}} > \kappa_{\text{HET-LITE}} > \kappa_{\text{HET}}$) and the primary contrast $\Delta\kappa_{\text{HOM-HET}}$ (positive, CI excludes 0) hold at every temperature; absolute κ levels vary non-monotonically, so it is the *contrast*, not the absolute agreement, that is temperature-stable.

T	κ_{HOM}	$\kappa_{\text{HET-LITE}}$	κ_{HET}	$\Delta\kappa_{\text{HOM-HET}}$	95% CI
0.0	0.33	0.17	0.03	0.30	[0.16, 0.44]
0.7	0.67	0.38	0.16	0.51	[0.21, 0.78]
1.0	0.53	0.27	0.15	0.38	[0.19, 0.54]

never touch the frozen study) at $T \in \{0.0, 0.7, 1.0\}$ and recomputed κ and the primary contrast at each temperature (Table 8). Two things are stable and one is not. *Stable*: the ordering $\kappa_{\text{HOM}} > \kappa_{\text{HET-LITE}} > \kappa_{\text{HET}}$ holds at every temperature, and the primary contrast $\Delta\kappa(\text{HOM} - \text{HET})$ is positive with a 95% CI excluding zero at every temperature (0.299, 0.509, 0.385 at $T = 0.0, 0.7, 1.0$). *Not stable*: the *absolute* κ levels vary non-monotonically with T (e.g. κ_{HOM} ranges 0.33–0.67), because temperature shifts both within-cell concordance and the directional marginals (at $T = 0$ the homogeneous marginal is near-uniform across BUY/HOLD/SELL, lowering chance-corrected agreement despite greedy decoding; at $T = 0.7$ –1.0 it concentrates on HOLD). The takeaway is that our conclusion is a statement about the *contrast* between configurations, which is temperature-robust, not about any absolute agreement level, which is not. This subgrid is a documented pre-registration amendment and is analyzed by the same seeded estimator as the main study.

6 Discussion

Our central result gives quantitative teeth to an intuition: a homogeneous LLM committee is substantially an echo chamber ($\kappa = 0.55$), and cross-provider heterogeneity is what actually makes its members think independently ($\kappa = 0.22$), for a well-characterized $\sim 4.4\times$ cost. The provider-pair matrix shows this de-correlation is driven by provider identity rather than sampling noise.

Crucially, de-correlation is a *necessary but not sufficient* condition for ensemble benefit: independence only helps if the independent opinions are also informative. Our honest null on *average* accuracy, read alongside the correlated-error / monoculture findings of Kim *et al.* [1], cautions that buying disagreement does not automatically buy correctness. Yet our exploratory analysis of consensus *structure* (Sec. 5.7) suggests the premium is not valueless in this setting: homogeneous committees produce confident, unanimous, wrong calls at $1.84\times$ the heterogeneous rate, so the independence appears to buy tail-risk reduction even without a mean-accuracy gain—

a hypothesis we flag for confirmatory follow-up. For practitioners, the frontier is directly actionable: teams paying the heterogeneity premium should verify that the additional independence translates into a downstream objective (calibration, robustness, or reduced collective-error rate) for *their* task rather than assuming it.

Methodologically, the sampling-non-determinism trap (Section 5.4) is, we argue, of independent value: any study that measures LLM-ensemble agreement without controlling the sampling context risks a multiplicative over-statement.

7 Threats to Validity

Heterogeneity vs. capability confound. HOM uses five instances of a small efficient model, whereas heterogeneous configurations add larger, different-provider models; part of the measured de-correlation could in principle reflect capability differences rather than provider identity per se. Two pieces of evidence bound this concern. First, the provider-pair analysis (Fig. 4) is a *within-cell* comparison: for a fixed cell, all agents receive the identical prompt, yet same-source agent pairs agree far more (0.80–0.92) than cross-source pairs (0.43–0.63), indicating that source identity—not merely stochasticity or prompt difficulty—drives disagreement. Second, the effect is monotone and significant at *every* increment of heterogeneity (Table 2), including the HET-LITE→HET step, which adds provider diversity without a systematic capability jump. Third and most directly, our capability-matched control (Section 5.8) holds tier approximately constant and still yields $\Delta\kappa = 0.315$ (95% CI [0.288, 0.341]), $\approx 94\%$ of the full effect, so the de-correlation is provider-driven rather than tier-driven. The residual limitation is scope: the control contrasts two providers (OpenAI vs. Google), since no efficient-tier xAI model is currently exposed; a fully tier-matched three-provider control awaits such a model.

Single task domain. All results are for one directional financial-analysis task with a single prompt template; richer analyst tasks may exhibit different agreement structure, and we make no claim about LLM ensembles in general. Two factors bound this concern. First, the driving mechanism—higher within-provider than cross-provider agreement (Fig. 4)—is a property of the models, not of the finance framing, so we expect the *ordering* (though not the magnitude) to transfer. Second, we pre-register an out-of-domain replication as immediate future work: the identical pipeline (three configs, per-agent seeding, per-run-averaged κ) applied to a three-class classification task (e.g. sentiment) on a small grid, reported without new confidence claims, to test whether the HOM > HET-LITE > HET ordering is finance-specific. Finance is arguably a *hard* case for stability, given the regime shifts spanned by our dates, which makes the monotone de-correlation we observe a

conservative demonstration.

Model-version pinning. Providers update models; we mitigate by pinning exact strings and treating the hash-frozen dataset as the object of record.

Single-run collection. Confirmatory data are collected once; provider-side drift within the window is not averaged out.

Accuracy underpowered. The design is powered for κ , not accuracy; the accuracy proxy is descriptive with wide intervals and must not be read as an alpha claim.

Generalization. Findings pertain to these three ensemble recipes, five agents, and this provider set; other sizes, tasks, and mixes may differ.

8 Conclusion

We present a pre-registered, 54,000-call measurement of how cross-provider heterogeneity de-correlates LLM ensembles and what it costs. Heterogeneity reduces within-ensemble agreement monotonically— $\Delta\kappa = 0.336$ (95% CI [0.304, 0.369])—at a $\sim 4.4\times$ inference-cost premium, defining an explicit agreement–cost frontier. We contribute a measurement protocol that removes a $\sim 4\times$ inflation from shared-context sampling non-determinism, and we release a frozen, reproducible artifact. We find no reliable *average*-accuracy benefit at this scale, situating the result within the algorithmic-monoculture literature: diversity buys disagreement, but disagreement must still be shown to buy value. An exploratory analysis nonetheless indicates one concrete value—homogeneous ensembles commit confident, unanimous, directional errors at $1.84\times$ the heterogeneous rate—suggesting tail-risk reduction that a confirmatory study should test. A pre-registered, capability-matched control establishes that the effect is provider-driven rather than a capability-tier artifact ($\Delta\kappa = 0.315$, 95% CI [0.288, 0.341]); the remaining methodological gap is a fully tier-matched *three-provider* control, which awaits an efficient-tier model from a third provider. Two further steps follow: an out-of-domain replication to test whether the de-correlation ordering is task-general, and connecting the priced independence to downstream objectives (calibration, robustness, collective-error rate) so that it can be shown to buy value in this and other settings.

Data and Code Availability

To support reproduction, we release: (i) the immutable, read-only, SHA-256-stamped raw dataset (`runs.csv`; 54,000 logged calls with per-call model, seed, prompt hash, timestamp, tokens, cost, and raw response); (ii) the pre-registration and the cryptographic freeze receipt; (iii) the deterministic, seeded analysis pipeline that regenerates every figure and table

in this paper byte-for-byte, together with a replication check; and (iv) the orchestration harness including the HET-SAMETIER control configuration. Artifacts are archived at [Zenodo DOI, to be minted on acceptance] and mirrored at <https://github.com/<user>/price-of-decorrelation>. The dataset SHA-256 and commit hash are recorded in the manuscript’s supplementary manifest.

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References

- [1] J. Kim *et al.*, “Correlated errors in large language models,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2025. [Online]. Available: <https://arxiv.org/abs/2506.07962>
- [2] J. Wang *et al.*, “Mixture-of-Agents enhances large language model capabilities,” *arXiv:2406.04692*, 2024.
- [3] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, “Improving factuality and reasoning in language models through multiagent debate,” *arXiv:2305.14325*, 2023.
- [4] S. F. Tekin, F. Ilhan, T. Huang, S. Hu, and L. Liu, “LLM-TOPLA: Efficient LLM ensemble by maximising diversity,” in *Findings of the Assoc. Comput. Linguistics: EMNLP*, 2024. [Online]. Available: <https://arxiv.org/abs/2410.03953>
- [5] “DFPE: A diverse fingerprint ensemble for enhancing LLM performance,” *arXiv:2501.17479*, 2025. [Online]. Available: <https://arxiv.org/abs/2501.17479>
- [6] Z. Bugaud, “Hidden Clones: Exposing and fixing family bias in vision-language model ensembles,” *arXiv:2603.17111*, 2026. [Online]. Available: <https://arxiv.org/abs/2603.17111>
- [7] “Council Mode: A heterogeneous multi-agent consensus framework for reducing LLM hallucination and bias,” *arXiv:2604.02923*, 2026. [Online]. Available: <https://arxiv.org/abs/2604.02923>
- [8] J. L. Fleiss, “Measuring nominal scale agreement among many raters,” *Psychological Bulletin*, vol. 76, no. 5, pp. 378–382, 1971.
- [9] J. R. Landis and G. G. Koch, “The measurement of observer agreement for categorical data,” *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.
- [10] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall, 1993.

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